

Pipeline Design Concepts

Computing and Mathematics

May, 2026



Pipeline Design Concepts (A14165)

Short Title:	Pipeline Design Concepts	
Faculty	Science and Computing	
Department	Computing and Mathematics	
Cluster	Media Production	
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Credits	5	Level: Introductory

Description of Module / Aims

This module provides the students with an introduction to the practical processes for realizing and integrating pre-production assets within the production pipeline. It exposes the student to the concepts and themes of the creative process and the interdisciplinary nature of the workflow phases. The theories and processes underlying the creative process are examined, while emphasis is placed on the practical realization of concept assets through the production of a digital animatic.

Programmes

COMP-0589	BSc (Hons) in Creative Computing (WD_KCRCO_B)	1 / 2 / M
COMP-0589	BSc in Multimedia Applications Development (WD_KMULA_D)	1 / 2 / M

Indicative Content

- Story Telling: Writing; The Process of Script-writing; Elements of story telling; Narrative construction; Dialogue
- Character Design and Development
- The Language of Film, Cinematography
- Storyboarding: Fundamentals; Technical Structure; Components and Functions; Case Studies; Script to Storyboard Realisation
- Call Sheets: Purpose and Overview
- Animatic: Timing, Creation of a basic digital Animatic from existing assets
- Production: Scene Selection and Setup
- Digitisation Process

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Identify and recognise the conceptual themes relating to Story telling, Script-writing, Story-boarding and Character Development.
2. Identify and recognise the key elements of Cinematography for story telling.
3. Interpret and identify the Production Pipeline phases.
4. Complete an animatic using industry standard software.

Learning and Teaching Methods

- This module will be presented by a combination of lectures and practicals.
- The lectures will be used to introduce new topics and their related concepts.
- The practical element allows the student to put into practice the theoretical concepts covered in the lectures.

Assessment Methods

	Weighing	Outcomes Assessed
Final Written Examination	50%	1
Continuous Assessment	50%	
Portfolio	50%	2,3,4

Assessment Criteria

<40%: Unable to interpret and demonstrate key concepts of the production pipeline.

40%-49%: Be able to interpret and demonstrate key concepts in the production pipeline.

50%-59%: Ability to interpret key concepts of the production pipeline and ability to discover and integrate related knowledge in other knowledge domains.

60%-69%: Be able to solve problems using the production pipeline process skills by experimenting with the appropriate skills.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of both complex and unforeseen problems through the use and modification of appropriate skills.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Supplementary Material

- Dunlop, R. _Production Pipeline Fundamentals for Film and Games_. 1st ed. London: Focal Press, 2014.
- Glebas, F. _Directing the Story: Professional Storytelling and Storyboarding Techniques for Live Action and Animation_. 1st ed. London: Taylor and Francis, 2009.
- Moritz, C. _Scriptwriting for the Screen (Media Skills)_. 2nd ed. New York: Routledge, 2008.

Requested Resources

- Lecture Room: Loose Seated
- COMPUTER LAB: Multimedia Lab